

SIMATS ENGINEERING

C04- ASSESSMENT TOOL 2

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

NAME : K.PUJITHA

REG NO : 192311124

COURSE OUTCOME COVERED

CO4: Implement network security testing methods by performing web server attacks, analyzing web application vulnerabilities, and assessing wireless network security using Kali Linux.

QUESTIONS: 01

Total Marks:100

1. TJX Companies Wireless Network Breach, 2005–2007

TJX Companies experienced a major payment-card data breach after attackers obtained unauthorized wireless access to its retail network. Investigations found that TJX had used WEP while transitioning to WPA and had not implemented adequate controls to restrict wireless access. The incident demonstrated the risks of obsolete encryption, inadequate monitoring, weak network segmentation, and insecure transmission of sensitive data.

Assignment Question: Prepare a Vulnerability Assessment Report examining the wireless encryption, authentication, network segmentation, monitoring, and payment-data protection weaknesses that contributed to the TJX incident. Recommend how WPA2/WPA3, enterprise authentication, rogue-access-point detection, and PCI DSS controls could have reduced the risk.

a)

Step 1 – Host Discovery

nmap -sn 192.168.56.0/24

Step 2– Vulnerability Scan

nmap --script vuln 192.168.56.102



2. Lowe's Wireless Network Intrusion, 2003

In 2003, attackers compromised the wireless network of a Lowe's retail store in Southfield, Michigan. The compromised wireless connection provided access to the company's central systems and other retail locations. The attackers attempted to install software intended to capture customers' credit-card information.

Assignment Question: Prepare a Vulnerability Assessment Report identifying the weaknesses that allowed a local wireless compromise to create access to a nationwide corporate network. Assess authentication, wireless isolation, firewall controls, point-of-sale segmentation, intrusion detection, and centralized access management.

A)

Step 1 – Port & Service Scan

`nmmap -sS -sV -p- 192.168.56.102`

Step 2 – Vulnerability Scan

`nmmap --script vuln 192.168.56.102`



3. Firesheep and Public Wi-Fi Session Hijacking, 2010 The Firesheep browser extension demonstrated how attackers connected to the same open wireless network could capture unencrypted session cookies and impersonate authenticated users. The event highlighted the danger of websites encrypting only the login page while transmitting later session information without full HTTPS protection.

Assignment Question: Prepare a Vulnerability Assessment Report for a public Wi-Fi provider

affected by Firesheep-style session hijacking. Assess open-network authentication, session-cookie protection, HTTPS enforcement, HSTS, VPN use, wireless client isolation, and user-security awareness.

A)

Step 1 – HTTP Security Headers

curl -I <http://192.168.56.102>



4. DarkHotel Attacks Through Hotel Wi-Fi Networks

The DarkHotel campaign targeted business executives and government officials using compromised hotel networks. Selected guests connecting to hotel Wi-Fi were presented with fraudulent software-update prompts that delivered malware. The campaign demonstrated the risks associated with third-party Wi-Fi management, captive portals, insecure update mechanisms, and insufficient guest-network monitoring.

Assignment Question: Prepare a Vulnerability Assessment Report for the affected hotel environment. Examine guest authentication, captive-portal security, DNS integrity, network segmentation, third-party access, digital-signature verification, malware detection, and monitoring of privileged hotel-network systems.

A)

Step 2 – DNS

nslookup example.com 192.168.56.102



5. Marriott Wi-Fi Deauthentication Incident, 2014

The United States Federal Communications Commission investigated Marriott after equipment was used to send Wi-Fi deauthentication messages that prevented guests from using their personal hotspots. Marriott agreed to pay a civil penalty of \$600,000. The incident showed how unprotected wireless management frames could be misused to disconnect legitimate users.

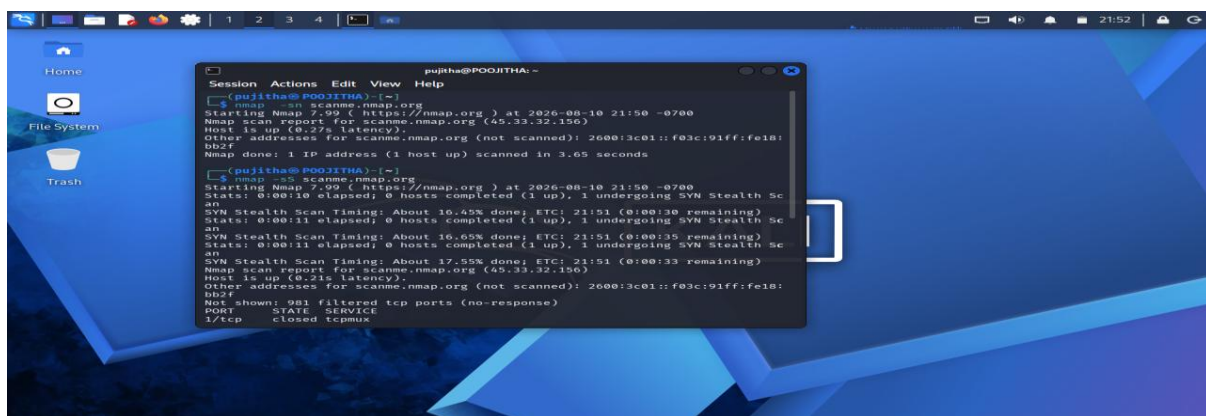
Assignment Question: Prepare a Vulnerability Assessment Report analysing the misuse of IEEE 802.11 deauthentication frames. Discuss availability risks, wireless interference,

Protected Management Frames, WPA3 requirements.

A)

STEP 1

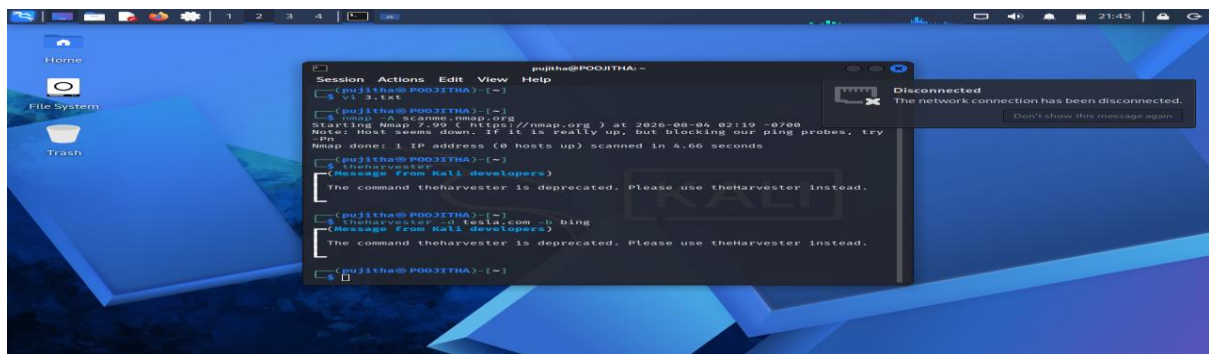
`iw phy | grep -i "management"`



6)

Step 1

`nmap -sV 192.168.56.102`



9. Australian Airport and In-Flight Evil-Twin Case, 2024

The Australian Federal Police charged a man who allegedly established fake free Wi-Fi access points that imitated legitimate networks at airports and other locations. The alleged purpose was to capture personal data from users who connected to the fraudulent access points.

Assignment Question: Prepare a Vulnerability Assessment Report for an airport or airline affected by an evil-twin access point. Analyse SSID impersonation, captive-portal authentication, automatic connection, certificate warnings, passenger-data exposure, rogue-access-point detection, staff response procedures, and user-verification methods.

Step 1

`sudo nmap -sn 192.168.56.0/24`



10. Network Rail Public Wi-Fi Portal Compromise, 2024

On September 25, 2024, public Wi-Fi services at 19 major United Kingdom railway stations displayed unauthorized offensive content. Reports stated that the Wi-Fi landing page was changed using a legitimate administrator account. The affected services were suspended while the incident was investigated.

Assignment Question: Prepare a Vulnerability Assessment Report examining the authentication and access-control weaknesses that may allow misuse of a legitimate administrator account. Assess privileged-access management, multifactor authentication, captive-portal security, third-party supplier access, logging, change approval, account monitoring, and incident-response procedures.

A)

Step 3

sudo nmap -sn 192.168.56.0/24

